

# Bihar Engineering University, Patna

## B.Tech 5<sup>th</sup> Semester Examination, 2025

Course: B.Tech  
Code: 100502

Subject: Control Systems

Time: 03 Hours  
Full Marks: 70

### Instructions:-

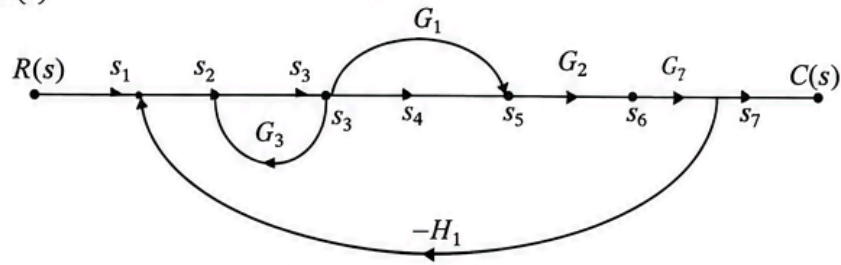
- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

### Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) A control system with feedback generally has
  - (i) Lower stability
  - (ii) ☒ Reduced steady-state error
  - (iii) Increased nonlinearity
  - (iv) Higher sensitivity to parameter variations
- (b) In signal flow graph analysis, a loop in contact with another loop is called
  - (i) Forward path
  - (ii) Self-loop
  - (iii) Non-touching loop
  - (iv) ☒ Touching loop
- (c) For a second-order underdamped system, increasing damping ratio decreases
  - (i) Rise time only
  - (ii) ☒ Peak overshoot
  - (iii) Settling time only
  - (iv) Natural frequency
- (d) The number of branches in a root locus equals the number of
  - (i) Zeros
  - (ii) Closed-loop poles
  - (iii) ☒ Open-loop poles
  - (iv) Gain constants
- (e) Breakaway points in root locus occur where
  - (i) Gain becomes zero
  - (ii) ☒ Multiple roots exist
  - (iii) Poles cross imaginary axis
  - (iv) Characteristic equation changes sign
- (f) A Type-1 system has zero steady-state error for
  - (i) ☒ Step input
  - (ii) Ramp input
  - (iii) Parabolic input
  - (iv) Impulse input
- (g) The slope of magnitude plot for a single pole in Bode plot is
  - (i) +20 dB/decade
  - (ii) ☒ -20 dB/decade
  - (iii) -40 dB/decade
  - (iv) +40 dB/decade
- (h) Phase margin is the additional phase lag required to make the system
  - (i) Critically damped
  - (ii) Oscillatory
  - (iii) ☒ Unstable
  - (iv) Faster
- (i) Root-locus based controller design mainly involves
  - (i) ☒ Pole-zero locations
  - (ii) Sampling frequency
  - (iii) Input amplitude
  - (iv) State variables
- (j) State-space analysis is particularly useful for
  - (i) Nonlinear algebraic equations
  - (ii) Single-input single-output systems only
  - (iii) ☒ Multi-variable systems
  - (iv) Frequency response only

- Q.2** (a) Explain signal flow graphs and derive Mason's Gain Formula. Determine the transfer function  $\frac{C(s)}{R(s)}$  for given signal flow graph. [7]



- (b) Discuss the effects of feedback on stability, sensitivity, bandwidth, and steady-state error of a control system. [7]

- Q.3** (a) A unity feedback system has: [7]

$$G(s) = \frac{25}{s(s+5)}$$

Determine (i) Damping ratio (ii) Natural frequency (iii) Peak overshoot (iv) Settling time.

- (b) Derive expressions for rise time and maximum overshoot for a standard second-order system. [7]

- Q.4** (a) Using the Routh-Hurwitz criterion, determine the stability of the system whose characteristic equation is: [7]

$$s^4 + 2s^3 + s^2 + 2s + 1 = 0$$

- (b) Sketch the root locus for:

$$G(s)H(s) = \frac{K}{s(s+2)(s+4)}$$

Determine the breakaway points and imaginary-axis crossings. [7]

- Q.5** (a) Explain the effect of adding poles and zeros on the transient response and stability of a control system. [7]

- (b) State and explain the Nyquist stability criterion. Also draw the Nyquist Plots for:

$$G(s)H(s) = \frac{3}{(s+2)(s-1)}$$

- Q.6** (a) Discuss gain margin and phase margin and explain their significance in relative stability analysis. [7]

- (b) Explain lag compensation and discuss its effect on steady-state accuracy and bandwidth. [7]

- Q.7** (a) Discuss the concept of robustness in control systems and explain how controller design improves robustness. [7]

- (b) Explain the concepts of state variables, state equations, and state-space representation with suitable example. [7]

- Q.8** (a) Discuss the stability of discrete-time systems and explain the role of eigenvalues in determining system stability. [7]

- (b) Test the controllability and observability of the system:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 0]$$

- Q.9** Write short notes on *any two* of the following: [7 x 2]

- Routh-Hurwitz stability criterion and its significance
- Gain margin and phase margin in frequency response analysis
- Initial value theorem and final value theorem
- Closed-loop control systems